

COURSE SYLLABUS

Linux Administration Essentials

COURSE NUMBER

LNK-101

DURATION

1 day · 4 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This foundational course provides hands-on instruction for administering Ubuntu Linux environments in enterprise settings. Students will master CLI navigation, user account administration, storage configuration, basic networking, and fundamental system security. Through interactive lessons and realistic lab scenarios, participants acquire the essential skills needed to manage and secure modern Linux infrastructure.

DELIVERY FORMAT OPTIONS

Standard Instructor-Led Training (ILT). The instructor teaches each module, leads explanations and discussions, and guides students through the standard module labs using the published module timing.

Bootcamp / Lab-Centered Delivery. When Bootcamp Mode capstone labs are included, they provide an alternative delivery configuration, not additional required ILT lecture time. In this configuration, students primarily work through module labs and capstone lab scenarios while the trainer acts as a facilitator, answering questions, clarifying concepts, and helping students work through lab challenges.

Training providers may choose either the normal ILT model or the Bootcamp/lab-centered model based on audience readiness, delivery goals, and available class time.

AT COURSE COMPLETION

After completing this course, you will be able to:

Manage local users, groups, and standard directory permissions safely on a Linux platform.

Initialize, format, mount, and diagnose problems with virtual block storage devices.

Formulate network settings, static IP allocations, and host resolution files on modern Ubuntu engines.

Secure host servers using service controls, packet-filtering rules, local audit logs, and elevation policies.

PREREQUISITES

- Basic understanding of operating system concepts
 - Familiarity with foundational networking terminology (IP addresses, subnets)
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COURSE OUTLINE

MODULE 1 User Account Management and File Permissions

105 min

Establish a strong administration foundation by mastering Ubuntu user provisioning, access control lists, and security-focused file permission structures.

LESSONS

- Lesson 1:** Introduction to the Ubuntu Shell and CLI Navigation
- Lesson 2:** Creating and Managing User Accounts
- Lesson 3:** Configuring Group Memberships and Permissions
- Lesson 4:** Applying Standard File and Directory Permissions
- Lesson 5:** Password Policies and Account Expiration

LAB EXERCISES

- Configuring Secure User Accounts and File Permissions on Ubuntu

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Navigate the Ubuntu command-line interface efficiently using core terminal shortcuts
- Provision new user accounts and user groups while maintaining administrative standards
- Modify standard Linux file permissions using symbolic and absolute modes

MODULE 2 Storage and File System Administration

105 min

Learn to manage storage devices, configure filesystems, and monitor disk resource constraints in an enterprise Ubuntu environment.

LESSONS

- Lesson 1:** Linux File System Hierarchy and Directory Structure
- Lesson 2:** Identifying and Managing System Hardware Devices
- Lesson 3:** Partitioning, Formatting, and Mounting Storage Volumes
- Lesson 4:** Monitoring Disk Space Usage and File System Health
- Lesson 5:** Managing Dynamic Storage and Logical Volume Management

LAB EXERCISES

- Storage and Filesystem Administration for Compliance Auditing

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Identify hardware specifications and partition configurations of connected storage drives
- Initialize and mount new file systems permanently via standard system configuration files
- Formulate command-line diagnostics to identify storage bottlenecks and low disk space conditions

MODULE 3 Network Configuration and Diagnostics

105 min

Acquire practical expertise configuring network interfaces, setting static and dynamic IP targets, and resolving connectivity bugs.

LESSONS

- Lesson 1:** Understanding Linux Network Interfaces and IP Addressing
- Lesson 2:** Configuring Static and Dynamic Networking with Netplan
- Lesson 3:** Essential Command-Line Network Diagnostic Tools
- Lesson 4:** Managing Hostnames, Name Resolution, and the hosts File
- Lesson 5:** Analyzing Network Connections and Socket Statistics

LAB EXERCISES

- Configuring Static IPv4 and Diagnosing Network Connectivity in Ubuntu

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Interpret network configuration files and interface properties using modern command structures
- Apply static and dynamic IPv4 network settings on an Ubuntu target machine
- Deconstruct connection failures using a systematic step-by-step diagnostic model

MODULE 4 Systemd Services, Logging, and Security Controls

105 min

Understand core system administration and security principles by managing services, tracking audit trails, and securing local networks.

LESSONS

Lesson 1: Understanding Systemd and Controlling System Services

Lesson 2: Restricting Privileged Access Using sudo and sudoers

Lesson 3: Securing Network Traffic with Uncomplicated Firewall (UFW)

Lesson 4: Auditing System Logs and Identifying Security Events

Lesson 5: Applying Essential Package Updates and Security Patches

LAB EXERCISES

Securing and Managing Nginx with Systemd and UFW

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Manage active Linux background tasks and system daemons utilizing standard service control actions

Enforce basic administrative isolation policies using sudo privilege elevation targets

Implement basic host-based traffic filtration strategies utilizing the local system firewall

BOOTCAMP MODE — CAPSTONE LABS

Capstone labs are included for Bootcamp/lab-centered delivery. They are not additional required lecture time for Standard Instructor-Led Training. Use them when the client chooses a facilitated lab delivery model where the trainer coaches students through integrated practice scenarios.

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

Create Linux users, establish group structures, and restrict file system access using numeric and symbolic permission masks

Provision new virtual disks, initialize filesystems, and configure persistent mount points in fstab

Implement static network configurations using Netplan and diagnose routing issues with command-line utilities

Deploy and secure system services, configure Uncomplicated Firewall (UFW) rules, and analyze syslog files for security events

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.